Geophys. J. R. astr. Soc. (1973) **34**, 27–46.

Palaeomagnetic Studies in the British Caledonides–III Igneous Rocks of the Northern Lake District, England

J. C. Briden and W. A. Morris

(Received 1973 March 19)*

Summary

Palaeomagnetic results are reported from the 'northern' Borrowdale Volcanics and from the Carrock Fell intrusive complex. The mean stable remanence direction in the volcanics is $D = 0°$, $I = -46°$ ($\alpha_{95} = 6°$) from 27 sites after allowing for tectonic dip. This result is supported by a decisive fold test in which precision improves by at least a factor of 3. The Carrock Fell Gabbro complex carries magnetization which is best interpreted as originating in Ordovician times when the complex was intruded as a dyke-sheet which has been subsequently only slightly warped. On this basis $D = 350°$, $I = -22°$, $k = 11{\cdot}7$ from 11 sites. A corollary of this interpretation is that the 'northern' Borrowdale Volcanics were folded before the emplacement of the Carrock Fell Complex. All these data are based on partial AF demagnetization. All sites have the same polarity. The corresponding palaeomagnetic pole positions are Lat. = 7° N, Long. = 177° E, $d\psi = 5°$, $d\chi = 8°$ and Lat. = 22° N, Long. = 188° E, $d\psi = 8°$, $d\chi = 15°$ respectively. These are in good agreement with other British Ordovician data.

1. Introduction

The English Lake District, a Lower Palaeozoic inlier, includes igneous rocks which are relatively undeformed in comparison with the rocks of the main geosynclinal belt within which it lies. Thus its structural situation is quite different from those of the rocks described in Papers I and II, which are marginal to the principal zone of subsequent orogenesis in the region. The inlier is basically a broad anticline with Cambrian–Lower Ordovician Skiddaw Slates forming the core. On the southern and shallower dipping limb the slates are overlain by over 3000 m of acid and intermediate lavas and pyroclasts referred to the Borrowdale Volcanic Series; these in turn are capped by late Ordovician and Silurian sediments (Fig. 1). By contrast on the northern limb of the major structure the Borrowdale Volcanic Series is thinner (~2500 m) and dips more steeply. There is nowhere continuity of outcrop of the Borrowdale Volcanics from the southern to the northern limb of the main anticline. This gives rise to many uncertainties in correlation which will not, however, concern us here because we are only concerned with the 'northern' Borrowdale Volcanics.

A preliminary investigation by Nesbitt (1967) suggested that rocks from Eycott Hill carried a stable remanence, with a direction similar to that in other British Palaeozoic rocks; but neither the sampling nor the stability evidence was adequate to

*Received in original form 1973 February 13

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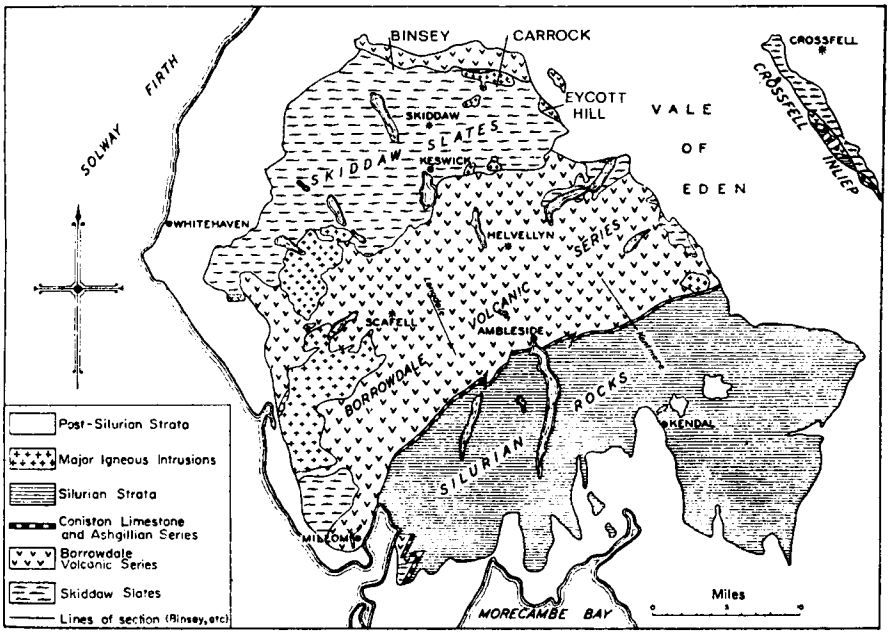


FIG. 1. Geological sketch-map of the English Lake District. Reproduced with additions from Eastwood (1953) by permission of the Director, Institute of Geological Sciences.

define the local Ordovician geomagnetic field satisfactorily. In this study comprehensive sampling was carried out mainly in two sections, at Binsey (15 sites) and Eycott Hill (15 sites), between which there is a large change of dip and strike. The results in this paper are supported by an elaborate fold test as well as by detailed laboratory evidence of stability.

The overall simple picture of Lake District geology is complicated by many large and small intrusions. The largest group comprises granites and granodiorites of

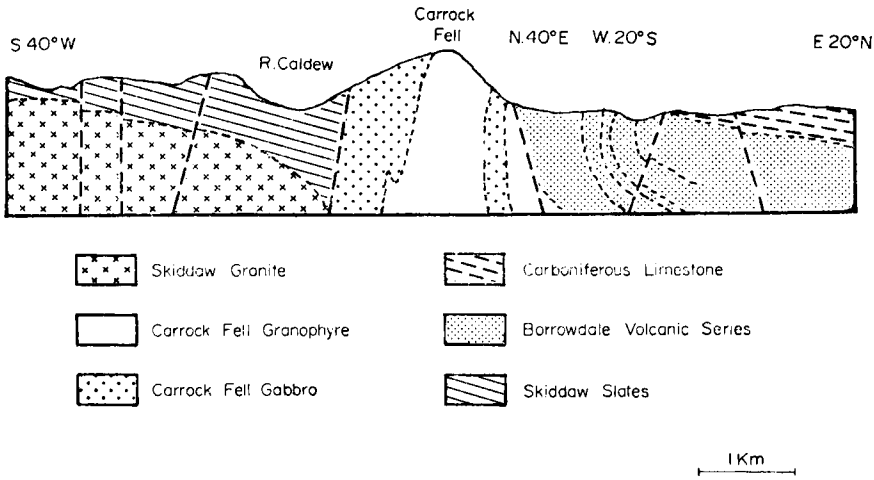


FIG. 2. Generalized geological section across the Carrock Fell Igneous Complex, simplified after the Geological Map Sheet 23, Cockermouth and Caldbeck, and reproduced by permission of the Director, Institute of Geological Sciences.

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Caledonian age, i.e. circa 400 My. One of the smallest outcropping granites, the Skiddaw Granite, has a metamorphic aureole which extends into the region sampled. There is also an older, more basic, suite of intrusives of which one of the most important forms the Carrock Fell Complex, which is intruded along the Borrowdale Volcanic/Skiddaw Slate junction close to the area where the lavas were sampled. The present field relationships are illustrated in the simplified geological section (Fig. 2).

The mode and time of intrusion of the Carrock Fell Complex in relation to the folding of its host rocks remains controversial. The extreme possibilities are injection as a 'sill' prior to folding, or post-deformational dyke emplacement (essentially in the configuration illustrated in Fig. 2). The age of the complex is likewise uncertain. The inclusion of xenoliths of Borrowdale Volcanics clearly defines a maximum age, and the overprint of the metamorphic aureole of the Skiddaw Granite provides a younger limit. It will be shown here that palaeomagnetism contributes to a more precise estimate of the age of the Complex as well as helping to determine the sequence of igneous and structural events.

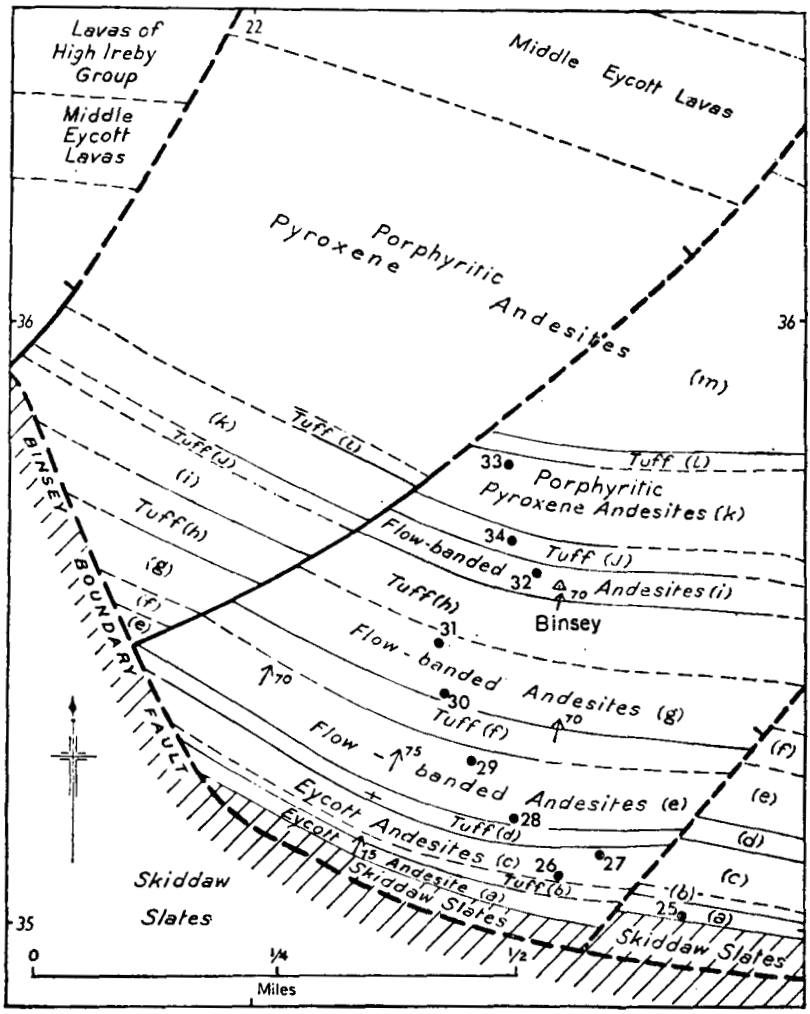


FIG. 3(a)

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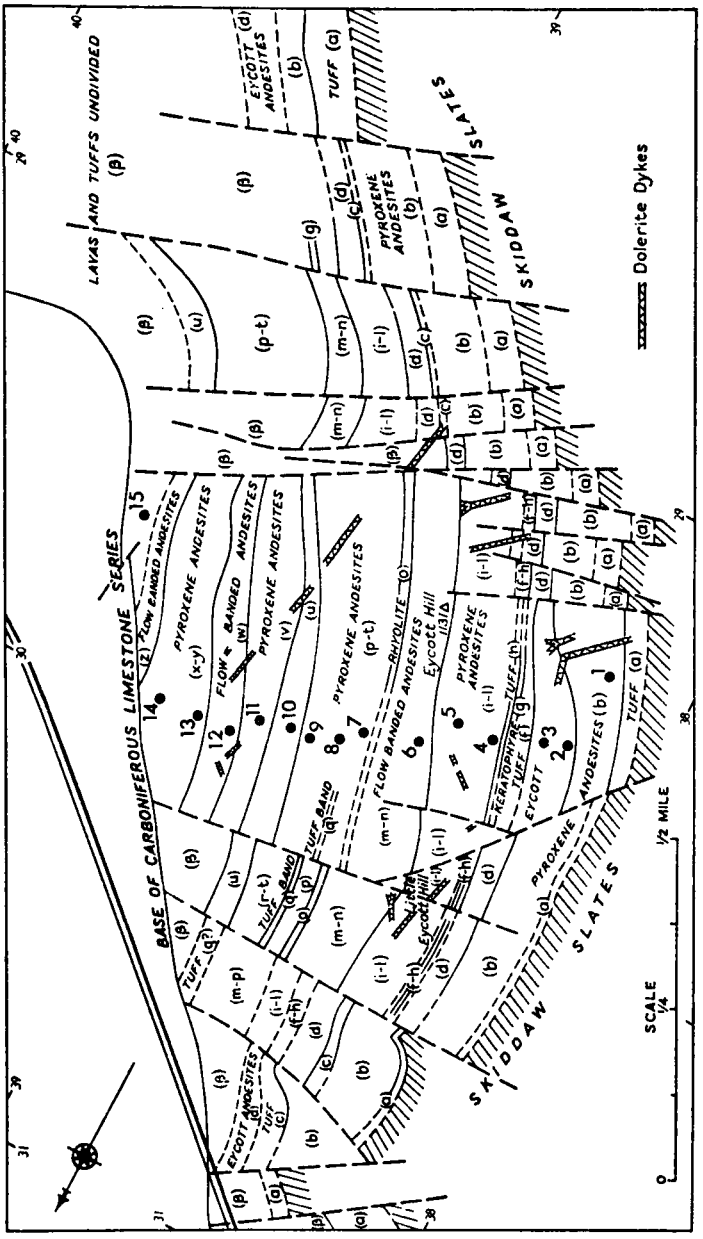

LAVAS AND TUFFS UNDIVIDED
BASE OF CARBONIFEROUS LIMESTONE SERIES
SKIDDAW SLATES
Dolerite Dykes
SCALE
½ MILE
0

FIG. 3(b)

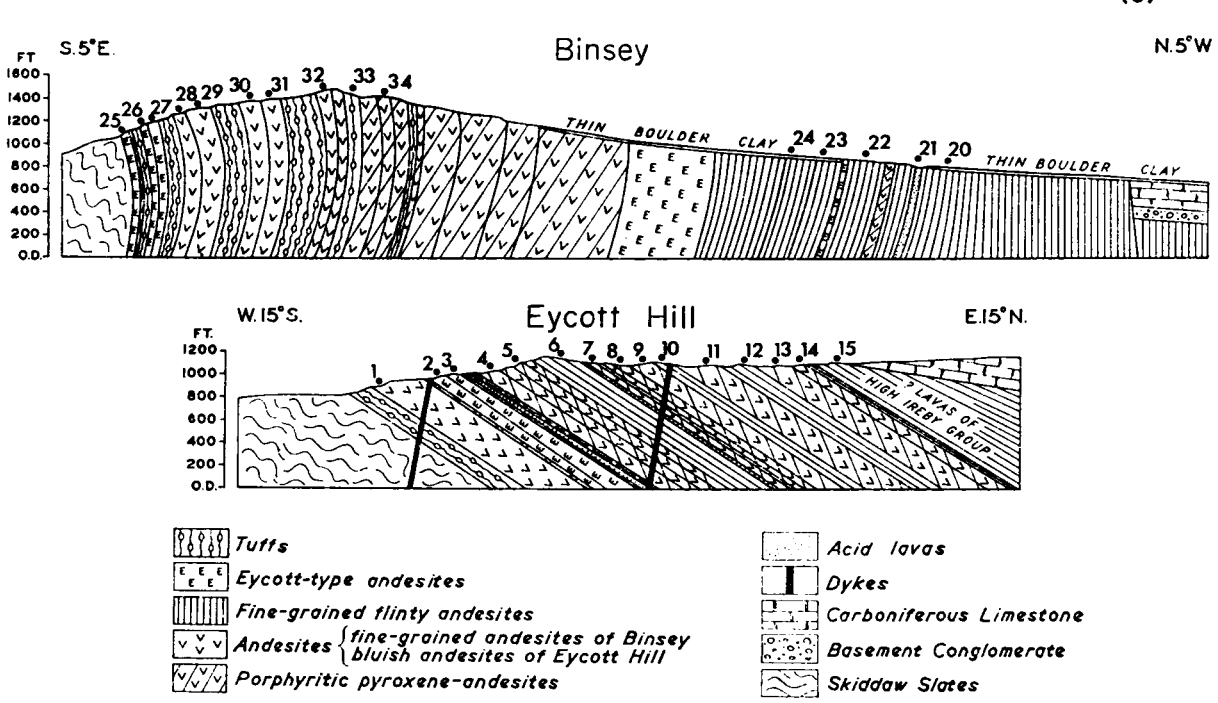


FIG. 3. (*a*) Geological map of the Binsey Group at Binsey. (*b*) Geological map of the area around Eycott Hill. (*c*) Schematic structural sections. Reproduced, with the addition of palaeomagnetic sampling localities, from Eastwood *et al.* (1968) by permission of the Director, Institute of Geological Sciences.

2. 'Northern' Borrowdale Volcanics

2.1. *Geology and sampling*

The earliest report on the northern area of Borrowdale Volcanics was by Ward (1877) under the title 'Lower Silurian lavas of Eycott Hill'. More recently the area has been comprehensively mapped by the (then) Geological Survey (Eastwood *et al.* 1968). Current geological interest in the region is concerned mainly with its detailed structural evolution.

The sequence consists of lavas, ashes, tuffs and agglomerates, ranging in composition from rhyolite to olivine-basalt, although the lavas are predominantly andesitic and the pyroclastics mainly acidic. The lavas are generally aphyric, but a distinctive plagioclase-phyric variety with phenocrysts up to 2 mm long—'Eycott-type' lava of Eastwood *et al.* (1968)—occurs at various stratigraphic levels, its occurrence eases the problem of correlation between the principal outcrop sections.

The series is divided into two groups, the Lower (Binsey) Group and the Upper (High Ireby) Group. Downie & Soper (1972) have suggested that these units should more properly be termed 'formations' and have proposed the name 'Eycott Group' to supersede the more vague 'northern Borrowdale Volcanics', but we have retained here the terminology of Eastwood *et al.* (1968). The Binsey Group is best developed in the west of the area notably on Binsey (Fig. 3(a) and (c)), where it attains its maximum thickness of 1190 m. The lowest beds are Eycott-type lavas (sites 25, 26) interbedded with mudstones and slates which are regarded as of *Didymograptus bifidus* age by Eastwood *et al.* (1968)—a correlation confirmed by the micropalaeontological data of Downie & Soper (1972). Ten sites in all were sampled in this type-section, of which nine are within a single fault block. Site 25 lies in an adjacent fault-block and continuity with the main section is not proven; but it will be shown later that these new palaeomagnetic results do not suggest any substantial age difference. The sequence thins drastically between Binsey and Eycott Hill so that, at the latter, the Binsey Group is only 120 m. Five sites were collected at Eycott Hill, where the oldest volcanic horizon (below site 1, Fig. 3(b)) is a red and green mottled tuff which corresponds to the level of site 34 on the upper southern slopes of Binsey. The base of the 'northern' Borrowdale Volcanics in the Binsey–Eycott area almost certainly falls in the *D. bifidus* zone, but is probably slightly younger within this zone on the eastern part of this area (Soper 1972, private communication).

The base of the High Ireby Group is marked by several flows by Eycott-type lava. As with the Binsey Group, its maximum development is near Binsey where it reaches about 1200 m. However on the sampled section at Binsey only the top 200 m are exposed (five sites, Fig. 3(c)); the intervening 1000 m to the top of the High Ireby Group being drift covered. At Eycott Hill the High Ireby Group is only 600 m thick but is well exposed as a result of strong glacial action. It is possible to distinguish separate lava flows, of which 10 were sampled. These may partially fill the gap in the record due to lack of exposure at Binsey. The eastward thinning of the Borrowdale volcanics can be demonstrated in each of the major stratigraphic subdivisions. Absolute thicknesses for the High Ireby Group, however, are additionally influenced by the overlap of the Carboniferous Limestone which terminates the Lower Palaeozoic outcrop at different stratigraphic levels along the angular unconformity. In the main section at Binsey the lavas dip steeply N at 70° while at the isolated site 25 the dip is only 30°. At Eycott Hill the sequence dips ENE at 38°. This difference in attitude between the two areas facilitates a fold test of the age of remanence. All collecting was by portable field drill and orientation by Sun and/or magnetic compass and clinometer.

2.2. *Results from the Binsey Group*

Total NRMs at 12 of the 15 sites are significant at the 95 per cent probability level

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(Table 1). The three sites with non-significant directions are in very fine grained rocks in which petrological alteration is visible in hand specimen. Ten of the site mean directions after dip correction are grouped around $D = 10°$, $I = -45°$. The other two are anomalous and relative to their *in situ* position, are grouped near the present geomagnetic field. The total NRM intensity ranges from $0{\cdot}05 \times 10^{-3}$ G to a maximum of $3{\cdot}0 \times 10^{-3}$ G.

Progressive AF demagnetization was performed on one specimen from each site.

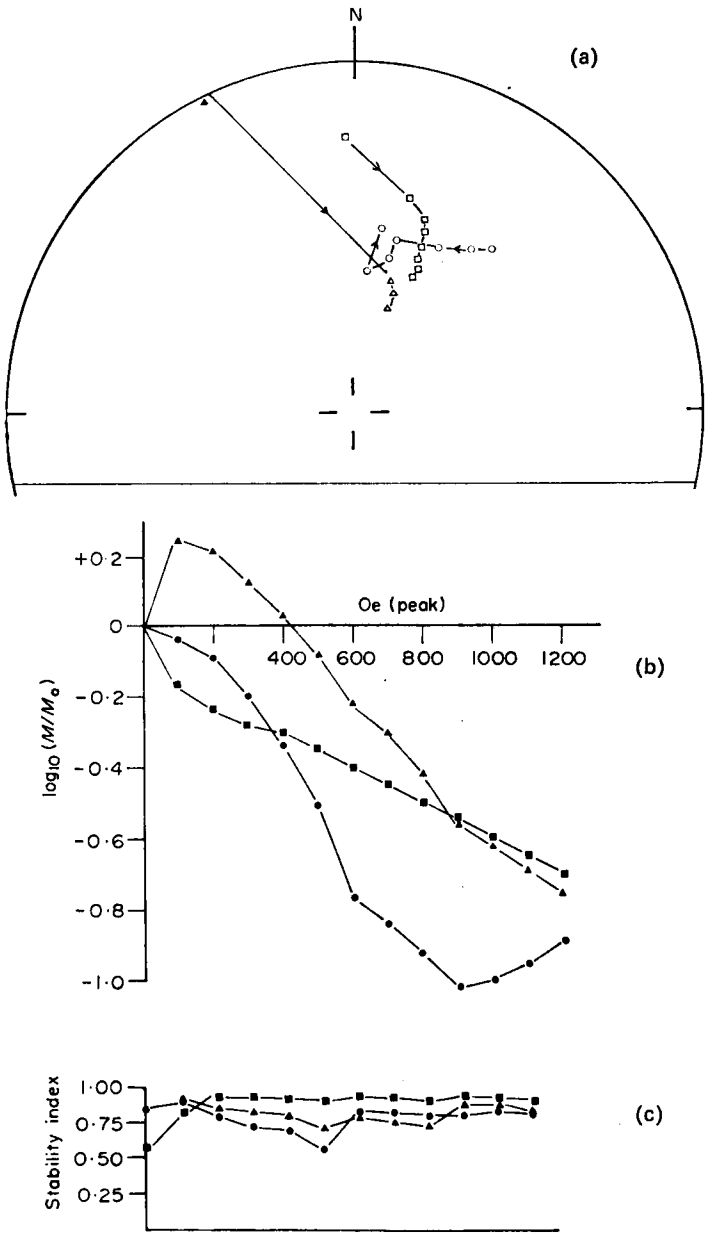


FIG. 4. AF demagnetization of specimens from the Binsey Group (*a*) variation of remanence direction, stereographic projection (throughout this paper open symbols denote upwards, and solid symbols downwards); (*b*) variation of intensity; (*c*) variation of stability index.

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Table 1

'Northern' Borrowdale Volcanic Series

Site	Location °N	°W	Total NRM N	R	k	α_{95}	*in situ* D	I	dip corrected* D	I	Peak field	AF cleaned N	R	k	α_{95}	*in situ* D	I	dip corrected* D	I	Virtual geomagnetic pole °E	°N
Binsey Group																					
25	54·7	3·2	6	4·85	4·3	36	0	+24	355	−42	200	6	5·92	62·0	9	3	−17	356	−45	181	+9
26			5	4·95	80·9	8	16	+5	11	−59	200	6	5·98	271·0	4	10	+9	352	−63	183	−9
27			6	3·37							200	5	4·47	8·0	30	357	+17	347	−46	189	+7
28			6	5·51	10·1	22	13	+16	8	−53	300	6	5·78	22·9	14	13	+14	359	−58	178	−4
29			6	4·68	3·8	40	5	+19	359	−48	200	6	5·93	72·5	8	3	+12	350	−54	185	+1
30			6	5·86	36·3	11	15	+44	16	−25	700	5	4·83	23·9	16	19	+20	19	−50	161	+3
31			6	5·91	55·5	9	16	+23	15	−47	500	6	5·92	61·9	9	16	+12	12	−58	168	−4
32			7	5·43	38·1	36	301	+82	13	+19	700	5	4·92	47·0	11	25	+26	26	−44	154	+6
33			6	4·48	3·3	44	5	+17	357	−50	700	6	5·94	304·0	4	21	+26	21	−44	158	+7
34			5	4·95	79·7	8	24	+32	24	−38	700	5	4·92	47·4	11	26	+30	26	−39	152	+10
1	54·6	3·0	6	5·66	14·9	18	13	+44	26	+13	500	6	5·82	28·8	13	20	+14	13	−25	163	+21
2			6	5·89	44·8	10	42	−4	37	−41	200	6	5·98	222·0	5	27	−32	358	−61	179	−6
3			6	3·32							900	6	5·44	9·0	24	355	−33	325	−43	209	+5
4			4	3·31	4·4	50	26	−25	5	−55	300	3	2·89	18·8	29	24	−20	6	−49	172	+5
5			6	3·34							700	6	2·76								
High Ireby Group																					
6	54·6	3·0	6	4·42	3·2	45	78	+13	78	−20	600	5	4·07	4·3	42	27	−32	358	−61	179	−7
7			6	2·96							600	5	4·09	4·4	41	40	−12	21	−47	159	+6
8			6	2·94							600	4	3·92	36·8	15	2	−18	344	−37	192	+14
9			6	5·19	6·2	29	34	+16	34	−19	700	5	4·42	6·9	31	30	−30	5	−61	174	−6
10			7	6·20	7·5	24	13	−11	0	−37	700	5	4·83	23·1	16	12	−10	0	−35	178	+16
11			6	5·44	8·9	24	10	−22	348	−44	100	6	5·86	36·7	11	9	−21	347	−42	189	+10
12			6	5·77	21·4	15	51	+12	51	−26	700	6	5·87	39·2	11	5	−17	347	−37	190	+14
13			6	5·86	35·9	11	40	+57	47	+20	700	6	5·88	43·0	10	17	+3	12	−26	165	+21
14			6	3·88	2·4	56	11	−16	353	−42	800	5	4·88	34·0	13	10	−17	353	−40	184	+13
15			6	5·44	8·9	24	42	+5	40	−32	800	5	4·83	24·2	16	14	−22	356	−44	180	+10
20	54·7	3·2	6	5·67	15·3	18	351	+43	358	−21	500	6	5·59	12·2	20	358	+31	356	−35	181	+16
21			6	5·33	7·3	26	355	+36	358	−29	200	5	4·95	94·0	8	356	+18	352	−38	185	+13
22			6	3·98	2·5	54	346	+34	349	−26	200	6	5·96	130·0	6	349	+34	352	−38	186	+20
23			5	2·54							700	5	2·62								
24			5	2·88							700	4	2·33								

* by simple rotation about the generalized local strike.

The behaviour of the majority of the samples can be categorized into two distinct groups:

(1) Little change of direction; regular decrease of intensity; uniform Stability Index SI (Briden 1972) (Fig. 4). This is identified as a single component remanence with a wide range of coercivities.

(2) Large change of direction in low alternating fields; large irregular decrease in intensity; and irregular fluctuations of SI (Fig. 4). This may be the result of a secondary magnetization superimposed upon a stable component with a limited range of coercivities. Fig. 4 also shows an exceptional demagnetization curve which exhibits a marked increase in intensity (twice NRM) upon 100 Oe treatment. Further treatment produces a gradual decrease of intensity, with constant direction and SI. Hence the initial step removes a weak secondary component of low coercivity.

Optimum fields for AF cleaning, selected according to the SI, ranged from 200 to 900 Oe. After this treatment directions at 14 of the 15 sites were significant. Two of the previously non-significant site directions (3, 27) became significant after cleaning at 900 Oe and 200 Oe respectively; site 5, however, failed to respond to treatment. Within-site grouping improved at most sites; only at sites 30 and 34 did precision decrease, and even there not by a significant amount by the criteria of McElhinny (1964) and Cox (1969). The mean directions at sites 1 and 32, which were anomalous before cleaning, now fall within the group centred at $D = 5°$, $I = -50°$ (after dip correction), with $k = 28{\cdot}7$, $\alpha_{95} = 8°$.

2.3. *Results from the High Ireby Group*

Eleven site mean directions out of 15 were significant prior to AF cleaning (Table 1). Within site grouping is poor ($k > 20$ only for sites 12 and 13). Two of the four non-significant site mean directions (23 and 24) were derived from more acid rocks than the others and, in any case, the samples were rather weathered; the other two sites (6 and 7) are in unaltered andesitic lavas. The total NRM results are grouped round $D = 20°$, $I = -29°$ after dip correction. Site 13 is an exception, the *in situ* direction being close to the present geomagnetic field. The intensity ranges from $0{\cdot}08 \times 10^{-3}$ G to $3{\cdot}0 \times 10^{-3}$ G, very similar to the distribution of intensities in the Binsey Group.

Progressive AF demagnetization was applied (Fig. 5). For most sites, the optimum treatment was between 600 and 800 Oe. The same categories of AF demagnetization behaviour can be distinguished as for the Binsey Group and in addition a common feature in the High Ireby Group is a well-marked flattening in the demagnetization curve around 600 Oe, accompanied by a sharp rise in SI. With further treatment, direction, intensity and SI all remain relatively constant.

After cleaning only two of the fifteen site means (23, 24) are non-significant and these are the weathered ones. All other sites (except 20) show an improvement of within-site group upon cleaning, although it is still poor in a few cases (6, 7, 9). Between-site grouping improves from $k = 6{\cdot}5$ to $38{\cdot}5$ upon cleaning which is significant by the criteria of McElhinny (1964) and Cox (1969). The mean direction is then $D = 356°$, $I = -41°$ with $\alpha_{95} = 7°$.

2.4. *Discussion*

The results (Tables 1 and 2) are derived by a single tilt correction by rotation about the present strike. This gives stable remanence directions for the Binsey Group and for the High Ireby Group which are significantly different (Watson 1956). This discrepancy could possibly be a consequence of oversimplifying the fold history. There have been several suggestions that the underlying Skiddaw Slates have been subjected to more than one episode of folding (Simpson 1967; Helm & Roberts 1971;

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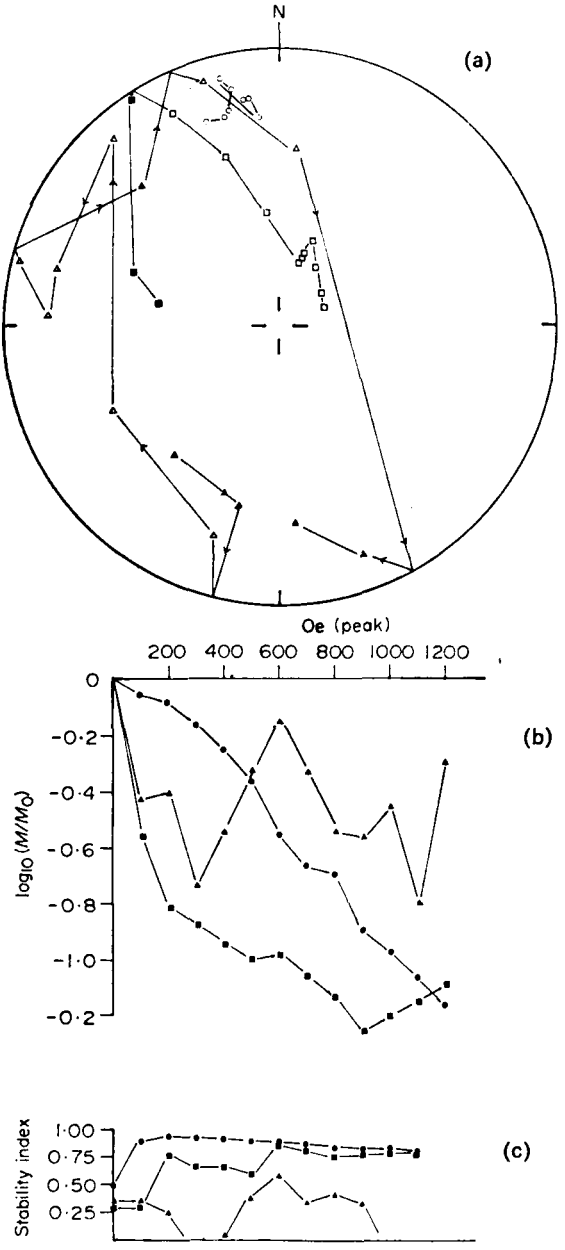


FIG. 5. AF demagnetization of specimens from the High Ireby Group, as Fig. 4.

Wadge 1971). Wadge (1971) contends that the Skiddaw Slates are folded by a major anticline plunging shallowly to the ENE. Helm & Roberts (1971) argue in favour of a steeper plunge at approximately 340° E of N. All these hypotheses envisage folding of the Skiddaw Slates before the onset of 'northern Borrowdale' vulcanicity, but, because it is likely that the Skiddaw Slate—northern Borrowdale volcanics sequence is conformable, it is worth testing the applicability of these hypotheses to the volcanics themselves. This has been done in the course of investigating the effects of *all* possible two-stage concentric deformations (Fig. 6) by allowing for each element in turn, starting with the younger. Accordingly Fig. 6 shows the results of first removing a

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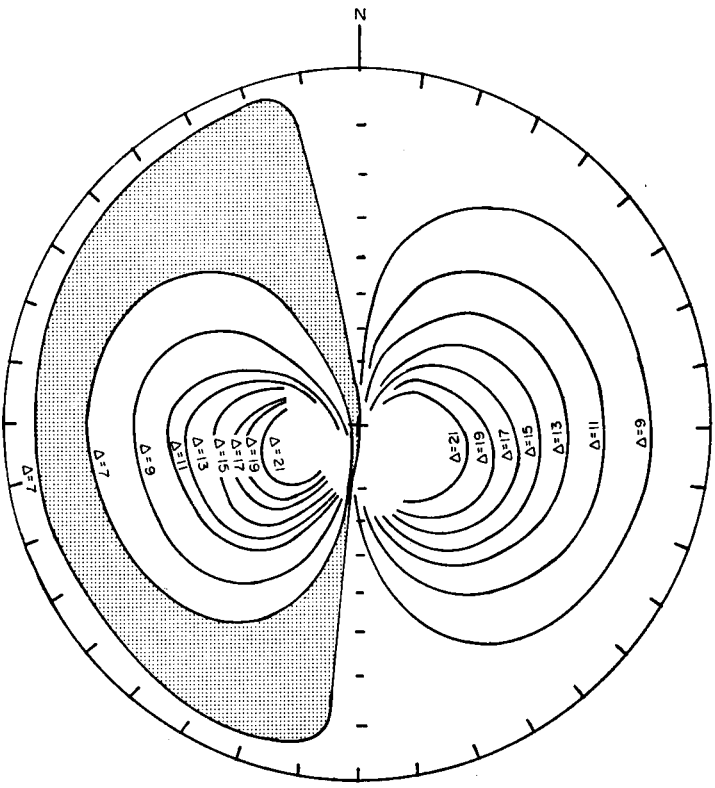


FIG. 6. Lower hemisphere equal-area projection of the angular difference (Δ) between the mean stable remanence directions in the Binsey and Eycott *sections*, assuming two elements of folding namely a plunge, and a residual simple dip. Δ is plotted as a function of assumed plunge. The zone of plunge corresponding to minimum Δ is stippled, but it should be noted that this minimum is very shallow and not significantly lower than the ' zero-plunge ' level.

plunge, and then allowing for the residual dip at each of our two sections. From this analysis, the optimum plunge correction would be taken as giving minimum angular discrepancy (Δ) between the results from the two sections.

Because of the uneven sampling of the two lava groups between the two sections it is necessary to treat the whole sequence together at each locality. In Fig. 6 the optimum plunge which might be inferred from the palaeomagnetic data is indicated by the stippled area, and it is evident that the plunge suggested by Helm & Roberts (1971) falls in this zone, though the plunge advocated by Wadge (1971) does not. However the Fisherian precision corresponding to this optimum hypothesis is 25·78 compared with 21·37 if zero plunge is assumed, and 19·50 on Wadge's hypothesis. These differences in precision are not significant with any high degree of probability, and the analysis merely *suggests* that Wadge's contention is less likely to be valid than the alternative hypotheses. Ramsay (1961) has shown, in the general context of structural geology, that failure to recognize plunges of 30° or less leads only to small errors in directional estimation (small, that is, by comparison with confidence limits commonly associated with palaeomagnetic results). Hence for palaeomagnetic purposes it is adequate to treat the deformation as a rotation or rotations about the present local strike at each locality, and the simplest interpretation would be of a single rotation about that strike although progressive tightening of this essentially simple structure is equally compatible with our evidence.

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Table 2

'Northern' Borrowdale Volcanic Series

							Palaeomagnetic pole			
Binsey Group	*N*	*R*	*k*	α_{95}	*D*	*I*	°E	°N	$d\psi$	$d\chi$
NRM *in situ*	12	10·59	7·8	16	15	23	155	46	9	18
NRM Dip corrected*	12	10·80	9·1	15	13	−37	164	14	10	18
AF cleaned *in situ*	14	12·89	11·7	12	14	6	159	37	6	12
AF Dip corrected*	14	13·55	28·7	8	5	−50	173	5	7	10
High Ireby Group										
NRM *in situ*	11	9·03	5·1	23	24	17	146	40	12	23
NRM Dip corrected*	11	9·46	6·5	19	20	−29	157	18	12	21
AF cleaned *in situ*	13	11·80	10·0	14	12	−8	104	31	7	14
AF cleaned Dip corrected*	13	12·69	38·5	7	356	−41	180	12	5	8
Combined										
NRM *in situ**	23	18·71	4·9	16	24	21	144	42	9	16
NRM Dip corrected	23	20·18	7·8	12	17	−33	160	16	8	13
AF cleaned *in situ*	27	24·55	10·0	9	13	−1	161	34	5	9
AF cleaned Dip corrected*	27	26·07	28·1	5	0	−45	176	9	4	7

* By simple rotation about the generalized local strike.

An F-test (Watson 1956) shows that the difference between the mean stable remanence directions in the Binsey and High Ireby Groups is significant. This could arise from a number of causes. First, an episode of deformation occurring between the formation of the two groups might be involved. A test of the kind exhibited in Fig. 6 is not sufficiently sensitive with the available data to detect such an event, but there is no geological evidence for its reality, indeed the dip and strike are remarkably uniform throughout each of the sections. Second, the significance of the difference might be a product of the statistical analysis. However the Watson & Irving (1957) analysis of dispersion, in the High Ireby Group in particular, shows that it arises mainly from within-site variation and the circle of confidence about the between-site mean is very small; hence the difference between the two groups is likely to be real. Third, even if real, the difference could be due to either genuine polar shift or failure to average secular variation. In Paper VI it will be pointed out that there is no evidence for systematic polar shift within the Ordovician of the British Isles. On the other hand, the possibility that secular variation has not been adequately averaged cannot be eliminated because in the absence of intercalated sediments, it is not known what time-span is covered by either of the two lava groups. Hence in presenting these results the data are combined for the whole of the 'northern' Borrowdale Volcanics in arriving at a best estimate of the local Ordovician palaeomagnetic field.

All samples from both sections exhibit the same polarity. The Builth Volcanic Series (Paper I) and the Aberdeenshire Gabbros (Paper II) have the opposite polarity and yet are dated at approximately the same age. In future, the construction of a polarity time-scale may help to define their relative ages more precisely.

Sites 5, 23, 24 were rejected from the final analysis because they gave random results. These sites are all close to the boundary between the Binsey and High Ireby Groups and the possibility arises that they may all have been formed at an 'instant' when the geomagnetic field was exceptionally weak. Although site 24 shows signs of incipient weathering and is more acidic in composition than most of the lavas, it is nevertheless petrographically very similar to site 21 which has a well-defined remanence, sites 5 and 23 are more basic fine grained rocks, showing no signs of weathering. The rocks are not petrographically distinct having approximately the same quantity and composition of magnetic mineral phases as the other rocks in the section. Nor are they magnetically distinctive in terms of remanent intensity or susceptibility. Hence the explanation of this possible 'randomly magnetized horizon' remains uncertain.

Site 25 is from the lowest lava in the Binsey Group and was collected in a small quarry at Whitfield Cottage. The lava is overlain by mudstones which pass upwards into subaerially deposited lavas—the 'Passage Beds' (Eastwood *et al.* 1968). Site 25 is separated from the main Binsey traverse by a N–S trending fault, and the dip is only 30° compared with the 70° dip of the main section. A simple tilt correction of 30° brings the remanence into line with those of all other sites in the 'northern' Borrowdale Volcanics. From this palaeomagnetic evidence there seems to be neither a substantial time interval nor an appreciable tectonic event between the formation of the lavas at site 25 and the next oldest sites (26 etc.). This supports geological evidence for stratigraphic continuity in the 'Passage-Beds' in this quarry, which is inferred from the absence of any discernable disconformity, and from micropalaeontological continuity (Downie & Soper 1972). Furthermore, sediment deposition was continuous at Eycott Hill throughout the time in question.

3. Carrock Fell Gabbro Complex

3.1. *Geology and sampling*

The Carrock Fell Complex is intruded along the complicated junction between the 'northern' Borrowdale Volcanics and the Skiddaw Slates to the south (Figs 1

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and 2). Initially Ward (1876) interpreted the complex as metamorphosed Borrowdale Volcanics. Harker (1894, 1895) remapped the area and recognized it as major intrusions of gabbro, diabase and granophyre. Subsequent mapping by the Geological Survey (Eastwood *et al.* 1968) confirmed this interpretation and introduced finer subdivisions.

The rock types are arranged in a series of steeply inclined sheets, running E–W parallel to the long axis of the Complex. Harker (1894), noting the compositional variation and distribution of rock types, suggested that the mass represented a single magma injection differentiated *in situ*. Eastwood *et al.* (1968) state that the ' several distinct types maintain their identities over considerable (lateral) distances and have fairly rapid transitional contacts '. This suggests deep-seated differentiation with pulses of magma intruded as dyke sheets. Dykes of similar composition and trend to the Gabbro Complex can be found in the surrounding country rock. The oldest intrusions within the Complex are the gabbros, beginning with a uniform melagabbro, and concluding with a leucogabbro. These were followed by the diabase, and the granophyre which occupies a central position within the Complex marks the last phase of intrusion. The age of intrusion is uncertain, it has been estimated by various workers as ' Tertiary, ' post-Silurian ' and ' Ordovician '. The only related radiometric data (Brown, Miller & Soper 1964) are on hornfels from the metamorphic aureole of the complex above Mosedale village, and give an age younger than the Skiddaw Granite which itself has a metamorphic aureole in the gabbro; therefore these data throw no light on the age of the Gabbro Complex. Exposure is best along the eastern margin where cliffs up to 30 m exist. Twenty one sites were collected by field drill. Sites were taken from the northern (2, 4, 5, 6) and southern margins, the Round Knott diabase (30, 31, 32) and from the metamorphic aureole of the Skiddaw Granite (10, 11, 12, 13, 14).

3.2. *Results from the Carrock Fell Complex*

The total NRM directions at only 14 of the 21 sites were significant ($p = 0{\cdot}05$, Table 3). Individual specimen directions show a distinct planar distribution near the N–S vertical plane (Fig. 7), which may indicate varying proportions of ancient remanence together with a secondary component along the present geomagnetic field. Because of the planar distribution no realistic mean direction can be determined. The exceptional range of total NRM intensities ($0{\cdot}3 \times 10^{-6}$ G to $7{\cdot}0 \times 10^{-3}$ G) presumably reflects the wide variety of rock types and, perhaps, the localized incidence of metasomatism.

The progressive AF demagnetization curves are mostly complex but three types can be identified:

(1) Sites 10, 11, 12, 13, 14, 31, 32. Direction and intensity change randomly during demagnetization while the Stability Index is erratic. Hence, it is not possible to select any optimum field for cleaning. Five of these seven sites (10, 11, 12, 13, 14) are from the metamorphic aureole of the Skiddaw Granite. Metasomatism, associated with the granite intrusion, which obviously bleached the gabbro, has probably altered both the abundance and composition of the magnetic fraction. Sites 31 and 32 from the Round Knott diabase also show this complex behaviour, possibly due to recent IRMs as these sites are from near the peak of Carrock Fell. These sites are magnetically unstable.

(2) Most sites (1, 2, 4, 6, 15, 16, 17, 18, 19, 30) have demagnetization curves which are much smoother than type 1, but they are still not simple. The most important characteristic is the stepped form of the Stability Index curve (Fig. 8)—steps at 200 and 800 Oe being most common. Intervals of little directional change with

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Table 3

Carrock Fell Gabbro Complex

Site	Location °N	Location °W	Total NRM N	Total NRM R	Total NRM k	Total NRM α_{95}	Total NRM *in situ* D	Total NRM *in situ* I	Peak field	AF cleaned N	AF cleaned R	AF cleaned k	AF cleaned α_{95}	AF cleaned *in situ* D	AF cleaned *in situ* I	Dip corrected* D	Dip corrected* I	Virtual geomagnetic pole Dip corrected* °E	Virtual geomagnetic pole Dip corrected* °N
2	54·7	3·1	6	3·00					300	6	3·51								
4			5	4·62	10·6	25	291	+32	700	6	5·31	7·3	27	345	−49	333	−52	200	0
5			6	5·21	6·3	29	6	+33	400	6	5·46	9·3	23	4	−1	3	−7	173	32
6			4	3·80	14·8	25	311	+73											
10			6	4·65	3·7	40	44	+72											
11			6	5·67	15·0	18	191	+10											
12			6	4·28	2·9	48	119	+81											
13			7	5·07	3·1	41	192	−10											
14			5	1·41															
15			6	5·54	10·9	21	346	−2	200	6	5·13	5·8	31	338	−21	334	−23	204	20
16			6	4·67	3·8	40	193	−30	300	6	5·50	10·0	22	351	−30	347	−34	190	16
17			6	3·30					200	7	5·46	3·9	35	323	−22	319	−21	219	16
18			4	2·52					400	6	5·84	30·3	12	351	−23	347	−27	190	20
19			5	2·41					200	5	3·93	3·8	46	22	+11	23	+3	149	34
21			6	5·78	22·8	14	23	−33	500	6	5·63	13·3	19	7	−35	1	−41	176	12
22			6	4·16	2·7	50	16	−10	400	6	5·89	46·0	10	8	−15	5	−21	171	24
23			5	4·99	448·6	4	349	−9	200	5	4·99	435·5	4	350	−14	348	−18	190	26
24			4	1·88					300	4	2·05								
30			6	5·65	14·2	18	287	+62	500	5	4·88	14·6	25	347	−27	344	+5	197	36
31			6	0·73					700	6	3·12								
32			6	5·25	6·7	28	163	+57											
AF cleaned 11 sites											10·15	11·7	14	353	−18	350	−22	188	28
																		($d\phi = 8°$, $d\chi = 15°$)	

* For tilt of nearby Carboniferous limestone.

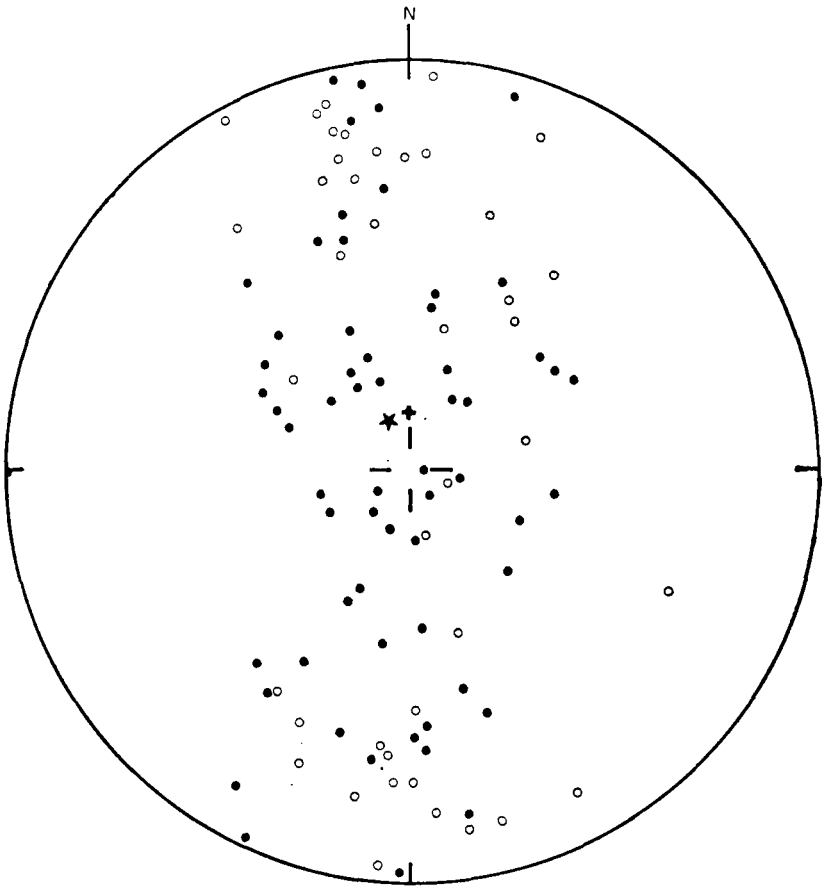


FIG. 7. Total NRM directions in specimens from the Carrock Fell Complex, showing planar distribution. The present geomagnetic field is denoted by a star and the geocentric axial dipole field by a cross. Well-grouped directions are obtained after AF cleaning, as discussed in the text.

progressive demagnetization are common, and coincide with the optimum SI; hence it is possible to select an optimum field for AF cleaning.

(3) The remaining sites exhibit a consistent SI, intensity falls exponentially and the direction remains fairly constant throughout treatment, close to the typical stable NRM for the whole complex. All the NRM in this group is likely to be original.

The response to cleaning was variable and 11 sites qualify for inclusion in the final analysis of the stable remanence. These all show the same polarity, which is also that of the 'northern' Borrowdale Volcanics. (That is not to imply, though, that the intrusives and extrusives necessarily originated in the same polarity epoch.)

4. Structural interpretation of palaeomagnetic results

In order to interpret the results from the Carrock Fell Complex it is necessary to find the mean remanence at the time of emplacement of the Complex so its deformation history must be determined. The *in situ* stable remanence ($D = 353°, I = -18°$, $\alpha_{95} = 14°$) is similar to the local Ordovician geomagnetic field direction, as inferred from other studies (see Paper VI). However this cannot be accepted as the optimum

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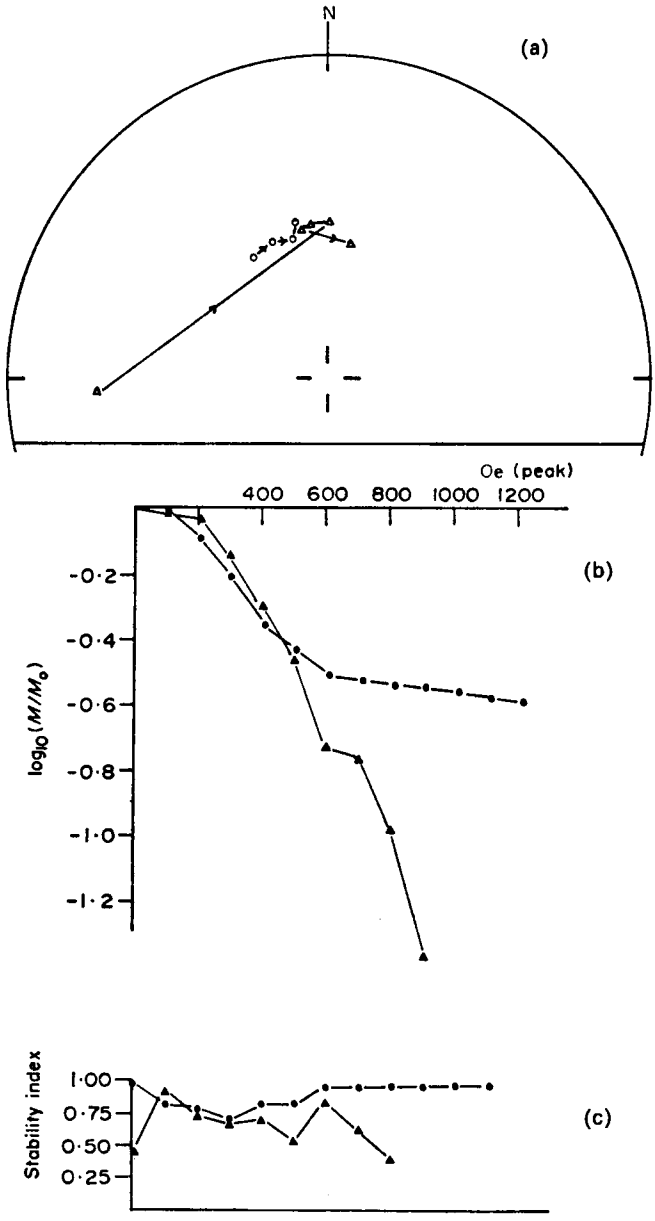


FIG. 8. AF demagnetization of specimens from the Carrock Fell Complex, as Fig. 4.

result because the shallow regional ENE dip of the Lower Carboniferous must necessarily be applied wherever a pre-Carboniferous magnetization is inferred. This tilt correction applied to the Carrock Fell results gives $D = 350°$, $I = -22°$, $\alpha_{95} = 14°$ which agrees even more closely with the direction inferred from other results of Ordovician age. It emerges that this is the best estimate of the original remanence of the Carrock Fell Complex because other structural complications, which might have been thought relevant, lead to less acceptable results.

If the Carrock Fell Complex is a funnel shaped layered intrusion, the mineral lamination would originally have been planar in any given sector. Tilt correction to

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the site-mean palaeomagnetic directions on the basis of the mineral lamination has been performed separately on the data from the southern and northern margins and from Round Knott. In all cases it leads to a decrease of precision from which it is concluded that this supposition is false. Alternatively the Complex may originally have been intruded as a sub-horizontal 'sill' which has subsequently been tilted by about 70° to its present attitude. The application of any large realistic tilt correction leads to mean remanence directions which are uninterpretable in terms of the Ordovician, or indeed any younger, geomagnetic field for the region.

Hence the palaeomagnetic results strongly favour the view that the Complex was intruded essentially in its present position, subject only to slight post-Carboniferous warping. Also on palaeomagnetic grounds an Ordovician age is suggested for the complex (Paper VI). It follows that end-Silurian deformation in the Lake District did not appreciably affect the attitude of this massive intrusion.

The precision ($k = 11{\cdot}7$) which is associated with this interpretation is low by comparison with the Binsey Group ($k = 29$) and High Ireby Group ($k = 38$) or with large gabbroic complexes, e.g. Bushveld, $k = 68$ (Gough & van Niekerk 1959); Stillwater, $k = 23$ (Bergh 1970); Freetown $k = 43$ (Briden, Henthorn & Rex 1971) One possible explanation is that distinct pulses of magma were intruded over a long time interval. The presence of chilled margins between successive sheets within the Gabbro Complex (Eastwood *et al.* 1968) suggests that the time between successive injections of magma was sufficient for the consolidation of the previous phase. It is, therefore, possible that the precision is a measure of Ordovician secular variation. A second plausible cause of dispersion would be that later intrusions of magma might distort previously intruded and consolidated material and that such distortions might not be accurately compensated by tilt correction.

The possibility of external tectonic disturbance of the Complex must also be considered because the Drygill Shales of Caradocian age which immediately overlie the Complex are deformed into a small anticline whose axis is parallel to the Complex and whose limbs dip at 40° and 70° to the N and S respectively. No structures of this magnitude have as yet been mapped within the Carrock Fell Complex, and there is no reason to suppose that they exist. The prevailing view is that the Drygill Shales are older than the Complex and are intruded by it. Hence their folding need not be matched by corresponding deformation of the Complex itself.

The uppermost 'northern' Borrowdale volcanics of Late Llanvirn age define a maximum age for the Carrock Fell Complex, but it is probably rather younger as time must be allowed for the intervening deformation. The differences between the Carrock Fell pole position and that obtained from Silurian rocks in the British Isles (Paper VI) indicates a minimum age for the Complex. The best estimate for the age of the Carrock Fell Complex, on purely palaeomagnetic grounds, is thus Caradocian or Ashgillian (∓ half a stage or less).

The principal contributions of this work to the geological evolution of the northern Lake District are, first, to demonstrate that the Carrock Fell Complex was intruded after the main folding of the 'northern Borrowdale volcanics' as suggested by Eastwood *et al.* (1968). This conclusion is independent of detailed arguments about the precise age of the intrusive complex. Secondly our evidence also demonstrates that both the major folding of the northern Borrowdale volcanics and the subsequent emplacement of the Carrock Fell Complex took place before the end of the Ordovician, i.e. rather earlier than most workers in that area seem to have thought.

Acknowledgments

This work was carried out while one of us (WAM) was supported by a research studentship from the National Environment Research Council. Our thanks are due to Professor R. L. Wilson for loan of field equipment, to Dr N. J. Soper for much

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advice on geological matters and for valuable criticism of the manuscript; also to Dr J. D. A. Piper for his contribution to the fieldwork.

Department of Earth Sciences,
Leeds University,
Leeds LS2 9JT.

W.A.M. also at
Department of Earth Sciences,
The Open University,
Walton Hall, Milton Keynes,
Buckinghamshire.

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